SURGSTREAM: A WORKFLOW-ANCHORED PROACTIVE STREAMING FRAMEWORK AND BENCHMARK FOR REAL-TIME SURGICAL VIDEO UNDERSTANDING

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ABSTRACT

Surgical video understanding is essential for intraoperative assistance such as workflow monitoring, instrument preparation and procedural documentation. Delivering such assistance in real time requires a model to jointly perceive the ongoing scene, remember past surgical events, and respond before clinically relevant moments pass. Existing surgical video-language models either operate offline on complete videos or narrate only the current frames, failing to retain procedural evidence and to decide when a response is required. In this work, we propose SurgStream, a workflow-anchored proactive streaming framework for real-time surgical video understanding. We first introduce Workflow-Anchored Memory (WAM) to record structured instrument–action–target observations and fold them into phase-level summaries, providing compact and queryable surgical memory. Deadline-aware Transition Learning (DTL) is then devised to supervise response states within clinically bounded windows, enabling timely responses before procedural deadlines. We further introduce SurgStream-Bench, comprising 11.7K streaming interactions across 14 tasks covering real-time perception, backward memory queries, forward proactive QA, proactive alerts and caption memory. Experiments on SurgStream-Bench show gains over state-of-the-art methods, especially in memory and forward proactive QA.

***Index Terms*—** Surgical video understanding, streaming video LLM, workflow memory, proactive response

1. INTRODUCTION

Real-time surgical video understanding plays a crucial role in intraoperative assistance [1, 2], such as phase-aware instrument preparation, procedural documentation and progress tracking. Such assistance involves understanding the current scene, recalling past events and delivering responses within relevant procedural windows. Figure 1 illustrates these interaction requirements and the temporal modes considered in this work. Surgical videos are long, visually homogeneous and densely structured by the workflow [3]. For time-sensitive assistance, delayed responses may lose their practical value, making real-time surgical interaction challenging.

Early surgical video-language models primarily focused on visual question answering on images or complete videos [4, 5, 6, 7]. Subsequently, general and medical vision-language models [8, 9, 10] achieved significant progress in visual understanding, and real-time surgical commentary [11] narrated the current frames on the fly. More recently, streaming video large language models (LLMs) [12, 13, 14, 15, 16] have enabled causal perception and proactive response on continuous video streams, and OneStreamer [17] further unifies perception, memory and proactive response in a single model.

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Despite this progress, existing methods are not explicitly designed around surgical workflows. Free-text memory does not explicitly preserve phase boundaries or structured event counts, making retrospective queries difficult to support reliably. Response supervision at a single annotated time [17] also leaves the interval of acceptable responses implicit. These limitations motivate workflow-structured memory, supervision within valid response windows, and evaluation of both answer content and timing.

To address these limitations, we propose SurgStream, which combines workflow-structured memory with response supervision within valid procedural windows. Specifically, we propose a Workflow-Anchored Memory (WAM) that writes a timestamped *Observe* record whenever the instrument–action–target state changes stably and folds the records of each finished phase into a *Summary*, so that retrospective questions are answered from compact text that grows with the number of phases. We further devise a Deadline-aware Transition Learning (DTL) that derives a clinical window for every response target from expert labels and redraws the target response time inside it in each epoch, teaching the model to respond as early as the evidence allows and before the deadline. Moreover, we construct SurgStream-Bench, a streaming surgical interaction benchmark of 14 tasks across five clinical scenarios and five temporal modes. Extensive experiments demonstrate that SurgStream outperforms competing methods, with particularly strong gains in memory and forward proactive QA.

2. METHODOLOGY

As shown in Fig. 2, SurgStream performs real-time surgical video interaction within a single streaming video LLM. Given a video stream $\{x_t\}_{t=1}^{T}$ arriving at 1 fps, the model reads at each step t the causal context $C_t = [M_t; V_t; D_t]$, consisting of the memory M_t, the recent visual window $V_t = \{x_{t-W+1}, \ldots, x_t\}$ and the dialogue D_t, and emits a control state $s_t \in \{$*Silence*, *Standby*, *Response*, *Observe*, *Summary*$\}$ followed by optional text. On this basis, WAM and DTL anchor what to remember and when to respond to the surgical workflow without new network modules, and SurgStream-Bench evaluates real-time surgical video interaction.

2.1. Workflow-Anchored Memory

To provide compact and queryable surgical memory, we propose WAM, which anchors both the write points and the content of memory to the surgical workflow instead of letting the model decide the boundaries of free-text captions. Let u_t denote the structured surgical state at time t, *i.e.*, the set of instrument–action–target triplets or present instruments. When u_t changes and remains stable for κ seconds, the model writes an *Observe* record $o_\tau = \langle t,\ u_t \setminus u_{t-1} \rangle$ at $\tau = t + \kappa - 1$, listing the newly appearing elements as "[mm:ss] instru-

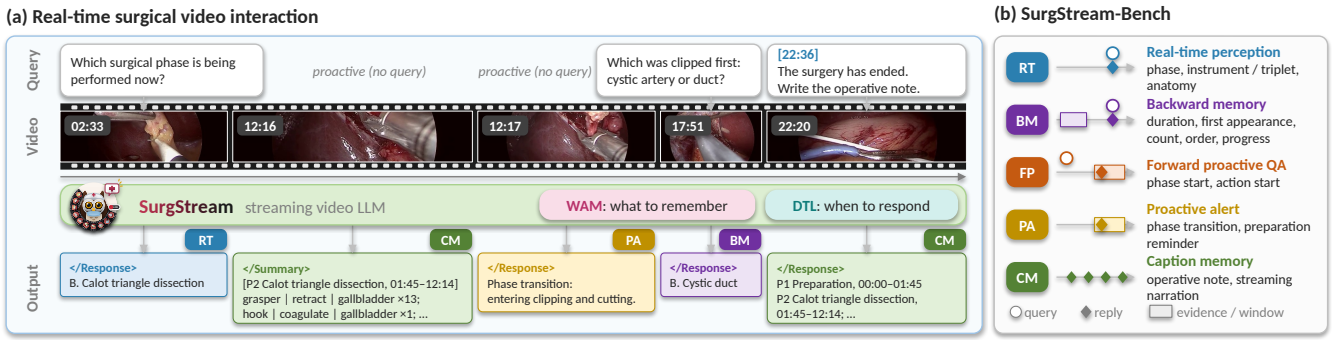


Fig. 1. Illustration of SurgStream and SurgStream-Bench. (a) An example of streaming surgical interaction, including current-scene questions, retrospective queries, proactive alerts and memory writing. (b) The five temporal modes evaluated by SurgStream-Bench.

ment | verb | target". After a phase boundary b_{k+1} is confirmed, the model writes a *Summary* record $\sigma_k = \langle p_k, [b_k, b_{k+1}), \{(c, n_k(c))\}\rangle$ for the finished phase p_k, where $n_k(c)$ counts the onsets of element c in the *Observe* records O_k of this phase (*e.g.*, clip×3), and folds these records into it:

$$M_{t+1} = (M_t \setminus O_k) \cup \{\sigma_k\}. \tag{1}$$

In this way, memory grows with the number of phases rather than with surgery duration, and questions on durations, first appearances and counts can be answered directly from the records. Since all records are derived from expert labels, memory writing is supervised by L_{mem}, the token-averaged cross-entropy of each record given its state and causal context. WAM thus provides queryable surgical memory that is verifiable against expert labels and remains compact for long procedures.

2.2. Deadline-aware Transition Learning

DTL supervises response states within clinical windows, extending generic sparse supervision [17] beyond a single annotated response time. Each response target j is assigned a window $[a_j, d_j]$, where the evidence time a_j and deadline d_j are derived from labels (*e.g.*, a preparation reminder is due at the next phase onset). To favour early responses, we resample the target time within the valid window at each epoch using inverse transform sampling. Let $U_j \sim \text{Uniform}(0,1)$ and $\Delta_j = d_j - a_j + 1$. We sample the response delay as

$$\tau_j = -\eta \log\left[1 - U_j\left(1 - e^{-\Delta_j/\eta}\right)\right]. \tag{2}$$

The response time is then

$$r_j = \min\{d_j,\ a_j + \lfloor \tau_j \rfloor\}. \tag{3}$$

Here, timestamps are discretized at one-second intervals, and $\eta > 0$ controls the decay of the sampling distribution. For $a_j = d_j$, we set $r_j = a_j$. We assign $s^*_{r_j}$ =*Response*, s^*_t =*Standby* for $t \in [a_j - \epsilon, r_j)$, and *Silence* elsewhere. Following [17], the supervised set Ω comprises all output anchors (including the resampled r_j), state transitions and representative state-persistence positions, and the state loss is

$$L_{\text{DTL}} = -\frac{1}{|\Omega|}\sum_{t\in\Omega} \log \pi_\theta(s^*_t \mid C_t). \tag{4}$$

Keeping Ω sparse prevents the abundant *Standby* positions from dominating the loss and suppressing *Response*. As r_j is redrawn in each epoch with earlier times favoured by Eq. (2), DTL teaches the model not only whether but also when to respond, encouraging earlier responses within valid windows.

SurgStream is trained jointly with the cross-entropy of the reply text, L_{mem} and L_{DTL}. At inference, the states are emitted in the order *Summary*→*Observe*→*Response* until *Silence* or *Standby*, and the context is bounded by the visual window and memory records, independent of surgery duration.

2.3. SurgStream-Bench

We construct SurgStream-Bench by converting expert annotations from Cholec80 [3] and CholecT50 [23] into streaming interactions. Phase, instrument and action-triplet labels are aligned along each procedure. Shared recordings are identified using the official overlap mapping [24], represented once with merged labels, and assigned to procedure-level splits to prevent leakage.

Following OneStreamer-1M [17], interactions cover real-time perception (RT), backward memory (BM), forward proactive QA (FP), proactive alert (PA) and caption memory (CM). Current-scene questions use labels at the query time. Retrospective queries derive durations, first appearances, counts and event order from earlier annotations. Phase intervals and triplet events provide summary and operative-note targets.

Proactive targets pair an expected answer with an evidence time and a deadline. Phase-transition alerts follow annotated transitions, while preparation reminders precede the next phase. This construction supports joint evaluation of answer content and response timing.

RT and BM use accuracy or task-specific tolerances. FP timing accuracy (T-Acc) checks the first reply after the query for correct content within $[a_j, d_j]$. PA timing F1 (T-F1) uses one-to-one temporal matching with the same requirements. CM uses structured F1 (S-F1) over phase intervals and triplet events. Exact vocabulary matching avoids an LLM judge. Avg_M and Avg_S average mode and scenario scores, each computed as the mean of its task scores.

3. EXPERIMENTS

3.1. Dataset

SurgStream-Bench uses Cholec80 (80 videos, 7 phases and 7 tools) and CholecT50 (50 videos with phase labels and 100 triplet classes), sampled at 1 fps. After resolving shared recordings, the training, validation and test splits contain 53/11/21 procedures spanning 32.5/6.1/15.5 h, respectively. The benchmark contains 11,667 interaction instances (7,315/1,320/3,032 across the splits), including 1,893 FP/PA instances. Its 14 tasks span five temporal modes and five clinical scenarios: workflow, instrument and action, anatomy, documentation and narration.

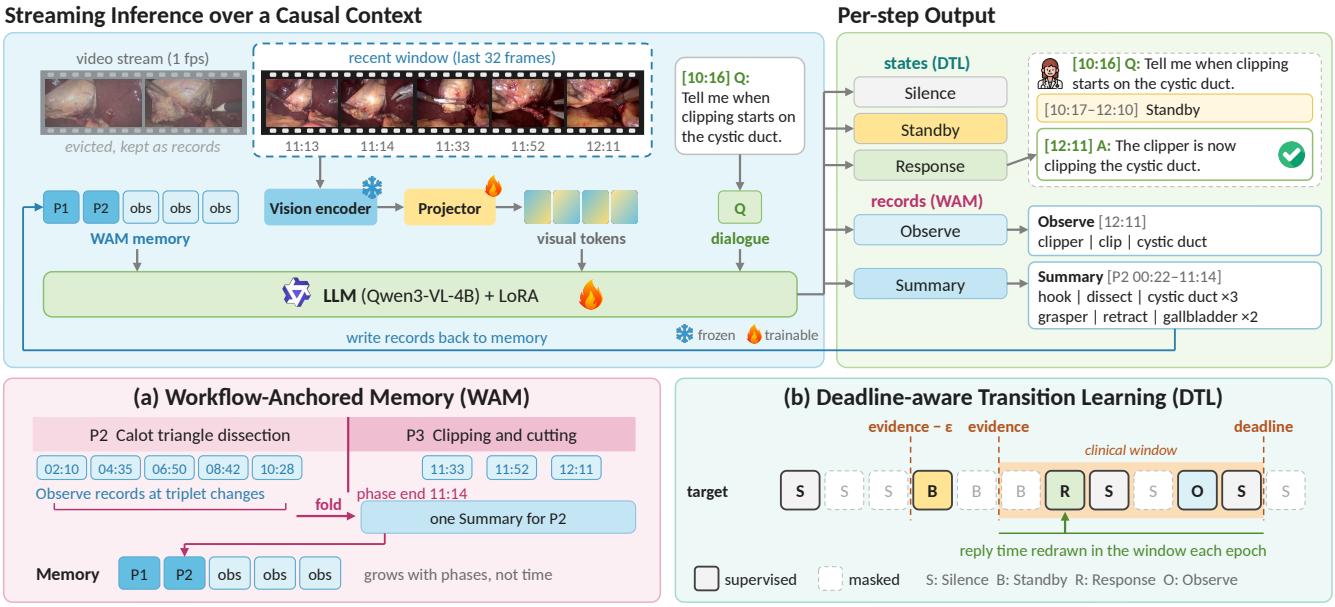


Fig. 2. Overview of SurgStream. At each step, the LLM reads WAM records, the recent visual window and pending dialogue, and emits a control state with optional text. (a) WAM writes *Observe* records at triplet changes and folds them into a *Summary* at each phase boundary. (b) DTL redraws the response time within the clinical window in each epoch and supervises states on a sparse mask.

Table 1. Comparison with state-of-the-art methods on SurgStream-Bench. †: fine-tuned on the SurgStream-Bench training split. The best and second-best results are in **bold** and underlined, respectively. All scores are reported in %.

Methods	Size	RT Acc↑	BM Score↑	FP T-Acc↑	PA T-F1↑	CM S-F1↑	Avg_M ↑	S1	S2	S3	S4	S5	Avg_S ↑
Qwen3-VL [8]	4B	41.73	29.93	0.43	0.30	4.26	15.33	12.11	30.13	46.95	0.68	8.52	19.68
Qwen3.5 [18]	4B	42.70	27.77	2.69	1.08	5.98	16.04	13.06	28.88	43.46	5.38	7.93	19.74
Gemma 4 [19]	E4B	42.89	19.88	0.86	1.60	6.62	14.37	14.17	13.06	52.39	4.64	9.08	18.67
MedGemma [20]	4B	34.54	11.14	0.43	0.73	3.48	10.06	8.84	11.04	36.66	0.89	5.17	12.52
Hulu-Med [10]	4B	42.77	29.70	0.00	0.78	5.95	15.84	13.18	30.15	42.23	6.01	7.28	19.77
MedVLM [21]	7B	55.35	27.80	0.54	1.84	2.43	17.59	13.75	33.36	53.13	0.68	4.85	21.16
EndoChat [22]	13B	39.62	27.92	0.00	0.21	0.86	13.72	5.29	29.57	55.37	2.33	0.48	18.61
OneStreamer† [17]	4B	83.27	59.10	10.87	14.78	15.16	36.64	32.99	62.72	74.81	18.49	24.47	42.70
SurgStream	4B	**84.07**	**64.13**	**23.13**	**16.64**	**25.22**	**42.64**	**36.84**	**67.70**	**78.06**	**31.45**	**28.79**	**48.57**

Table 2. Ablation study of SurgStream on SurgStream-Bench.

WAM	DTL	RT↑	BM↑	FP↑	PA↑	CM↑	Avg_M ↑	Avg_S ↑
		83.27	59.10	10.87	14.78	15.16	36.64	42.70
✓		82.74	62.46	14.53	17.58	22.84	40.03	46.74
	✓	83.48	61.97	20.23	14.37	22.34	40.48	45.96
✓	✓	84.07	64.13	23.13	16.64	25.22	42.64	48.57

Phase-transition alerts use $[b_k, b_k + 10\,\text{s}]$, and preparation reminders use $[b_k - 60\,\text{s}, b_k]$. Duration, first-appearance and count answers use tolerances of 20%, 10 s and ±1, respectively. Progress questions use S-F1. For CM, phase boundaries and triplet events are matched within 30 s, and narration is due within 10 s after each phase.

3.2. Implementation Details

Experiments run on RTX 6000 Ada and RTX 5090 GPUs. We initialize SurgStream from OneStreamer-4B [17] built on Qwen3-VL-4B [8]. As proactive targets and phase boundaries are rare, two extra training windows are centred on each FP/PA instance and each *Summary* write point. To mimic self-generated memory, the ground-truth memory prefix is perturbed by record dropping ($p = 0.1$), timestamp jittering (±2 s) and random element replacement ($p = 0.05$). We adopt two-stage LoRA fine-tuning: Stage 1 warms up memory writing for 1 epoch, and Stage 2 jointly trains all 14 tasks for 2 epochs. We use $W = 32$, $\kappa = 3\,\text{s}$, $\epsilon = 5\,\text{s}$, $\eta = 10\,\text{s}$, and apply LoRA with rank 16 and AdamW with an initial learning rate of 1×10^{-4}, cosine annealing and a global batch size of 32. States and text are decoded greedily. For fair comparison, OneStreamer† uses the same data, window sampling and epochs as SurgStream. Offline VLMs are evaluated in a pseudo-streaming manner by uniformly sampling up to 32 frames from the video start to the query and polling every 5 s with the last 8 frames for FP/PA.

3.3. Comparison with State-of-the-Art Methods

We compare SurgStream with general offline VLMs (Qwen3-VL [8], Qwen3.5 [18], Gemma 4 [19]) and medical VLMs (MedGemma [20], Hulu-Med [10], MedVLM [21], EndoChat [22]) in the zero-shot setting, and with the streaming video LLM OneStreamer [17] fine-tuned on the same data (OneStreamer†). As

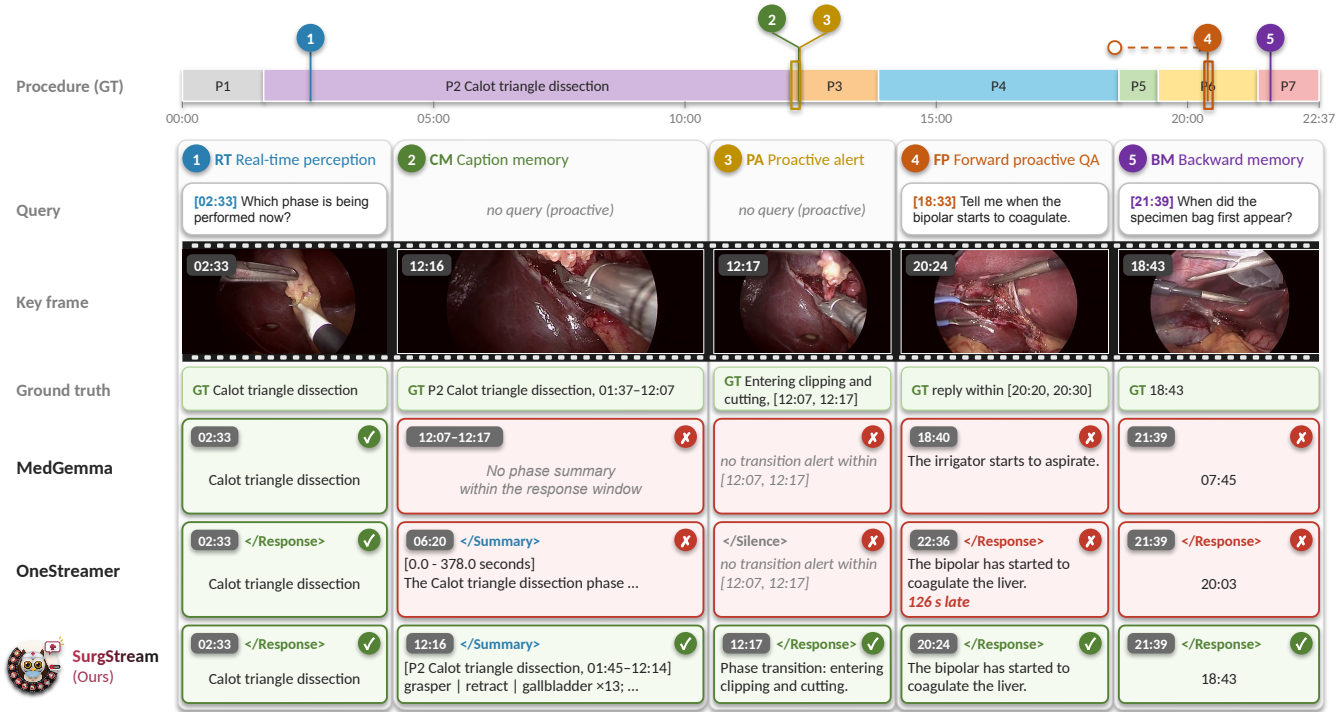


Fig. 3. Qualitative comparison of MedGemma, OneStreamer and SurgStream on one test procedure across five temporal modes. All three methods answer the RT query correctly. The remaining selected examples illustrate differences in memory and proactive responses.

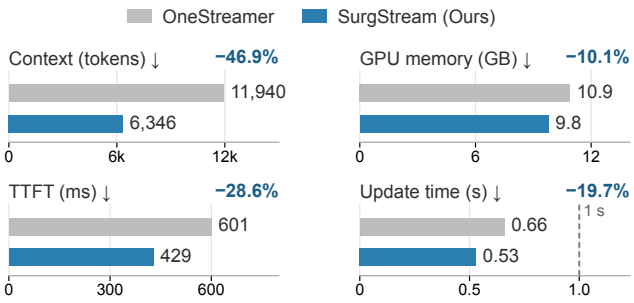


Fig. 4. Efficiency comparison with OneStreamer on full-length test videos at 1 fps on a single GPU. SurgStream reduces all four costs and keeps the per-step update below the 1 s input interval.

shown in Table 1, SurgStream achieves the best Avg_M of 42.64%, surpassing the second-best OneStreamer† by 6.00% (48.57% *vs.* 42.70% on Avg_S), with the largest gains on FP (+12.26%) and CM (+10.06%), highlighting the benefit of workflow-anchored memory and deadline-aware transition learning. Zero-shot VLMs stay below 3% on FP and PA, since offline models answer only when queried and cannot decide when to respond, and medical VLMs do not consistently outperform general ones (*e.g.*, MedGemma 10.06% *vs.* Qwen3-VL 15.33%). Across scenarios, SurgStream improves all five and gains most on documentation (S4, +12.96%), which relies on phase-level summaries. In Fig. 3, OneStreamer misses the phase-transition alert and answers the coagulation query 126 s late, whereas SurgStream recalls the specimen-bag appearance and responds within both proactive windows.

3.4. Efficiency Analysis

We measure context tokens (Ctx.), GPU memory (Mem.), time to first token (TTFT) and per-step update time (Upd.) on full-length test videos. In Fig. 4, SurgStream reduces context tokens by 46.9%, TTFT by 28.6% and GPU memory by 10.1% relative to OneStreamer. Its 0.53 s update time stays below the 1 s input interval. WAM provides these savings by replacing free-text captions with structured records that grow with the number of phases.

3.5. Ablation Study

We ablate both components on top of a baseline that uses free-caption memory and sparse state supervision (PSTL) [17]. This baseline is identical to OneStreamer† in Table 1, so all variants share the same data and training schedule. As shown in Table 2, introducing WAM improves BM by 3.36% and CM by 7.68%, DTL improves FP by 9.36% and CM by 7.18%, and combining both achieves the best Avg_M of 42.64% and Avg_S of 48.57%. Both components improve CM by over 7%. WAM brings the larger gain on BM and PA, whereas DTL brings the larger gain on FP. The combination performs best on all modes except PA, where WAM alone is higher (17.58% *vs.* 16.64%). Still, FP and PA stay below 24% for all variants, so response timing remains the main difficulty.

4. CONCLUSION

We introduced SurgStream, a framework combining workflow-anchored memory with deadline-aware response supervision for streaming surgical video understanding. On SurgStream-Bench, it improves memory-related performance and forward proactive accuracy while reducing context size and inference cost. Proactive response timing remains challenging, and the benchmark currently focuses on laparoscopic cholecystectomy. Future work will improve response timing and extend evaluation to other procedures.

5. COMPLIANCE WITH ETHICAL STANDARDS

This research study was conducted retrospectively using human subject data made available in open access by Cholec80 [3] and CholecT50 [23]. Ethical approval was not required as confirmed by the licenses attached with the open access data.

6. ACKNOWLEDGEMENTS

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